NRG1 fusion-hotspot benchmark report

Study: msk_impact_50k_2026

Abstract

This report analyzes NRG1 gene fusions from the msk_impact_50k_2026 study, covering 15 fusion events. 8/8 fusions with known frame status (100.0%) were in-frame. 13/15 fusions (86.7%) retained the Neuregulin intracellular region domain. Domain retention was tested with Fisher's exact test (p=0.227273, not statistically significant at alpha=0.05). No literature benchmark is configured for NRG1.

Results summary

Fusion partner frequency

Fusion-partner frequency was tabulated across 15 analyzed fusion events, identifying 6 distinct fusion partner genes. The most frequent partner was CD74, observed in 9 events.

Domain retention

Neuregulin intracellular region domain retention was tested with Fisher's exact test comparing in-frame fusions against all others; 8/8 (100.0%) of in-frame fusions retained the domain, p=0.227273 (not statistically significant at alpha=0.05). A breakpoint-position permutation test produced a corroborating empirical p-value of 0.732673.

Domain disruption

Disruption of the Immunoglobulin I-set domain and Human growth factor-like EGF was tested with Fisher's exact test comparing in-frame fusions against all others; 5/6 (83.3%) of in-frame fusions disrupted the domain, p=0.772727 (not statistically significant at alpha=0.05). A breakpoint-position permutation test produced a corroborating empirical p-value of 0.594059.

Junction concentration

100% of 8 fusions with this gene as the 3' partner join at exon 6, against 51% expected if breakpoints fell at random along the gene; this is the most over-represented junction (empirical p=0.043956, corrected for testing every exon, statistically significant at alpha=0.05). The random-breakage null assumes a breakpoint anywhere in the gene would be detected. Capture panels such as MSK-IMPACT tile only selected introns, so concentration on panel data is partly by assay design and this p-value overstates the evidence for biological concentration. Confirm in a cohort sequenced without intron selection (whole genome or RNA).

Cutpoint detection

Cutpoint detection scanned 12 mapped breakpoints for the protein position that best separates domain-retained from lost/disrupted fusions; the inferred cutpoint was position 1 aa (exon 1; label-permutation p=0.0909091). This boundary is descriptive: retained/lost status follows from breakpoint position, so a clean split exists by construction and the p-value is not evidence of selection. This is -40 aa from the nearest configured domain boundary at 41 aa.

Corroborating confidence statistics

Corroborating confidence statistics compared Domain_retention_flags groups "retained" (n=10) and "not_retained" (n=2). For Frame_status, retained: 6/6 (100.0%, 95% CI 61.0%-100.0%); not_retained: 0/0 (unavailable, 95% CI unavailable-unavailable). Welch's t-test comparing Tumor_variant_count between groups gave means of 27.4 vs 39.5, p=0.575883 (not statistically significant at alpha=0.05).

Composite evidence score

Composite evidence score reported: cutpoint_proximity sub-score excluded: corrected cutpoint p-value was missing or >= 0.05, cutpoint was not determinable, or mapped breakpoints were unavailable.

Exon retention

Exon retention produced results for this run (1 row(s) in Exon_retention); see the accompanying table.

Expression association

Expression association reported: No mRNA expression data was available for NRG1 in this cohort; expression-association analysis was skipped.

Genomic position recurrence

Genomic position recurrence produced results for this run (12 row(s) in genomic_bin_recurrence, 0 row(s) in exact_position_recurrence, 2 row(s) in protein_position_genomic_spread); see the accompanying table.

Joint partner

Joint partner reported: NRG1 has no gene_pair configured; joint-partner analysis was skipped.

Mechanistic interpretation

Mechanistic interpretation produced results for this run; see the accompanying data for details.

Mutation cooccurrence

Mutation cooccurrence reported: NRG1 has no mutual_exclusivity_targets configured; co-occurrence/mutual-exclusivity analysis was skipped.

Window detection

Window detection produced results for this run (14 row(s) in window_scan, 2 row(s) in top_windows); see the accompanying table.

Tables

composite_score tables

composite_evidence_ranking

Partner_gene	Event_count	Recurrence_score	Domain_retention_score	Domain_disruption_score	Cutpoint_proximity_score	Confidence_certainty_score	Components_applicable	Composite_score	Rank
CD74	9	0.6	0.01350896540516664	0.0226170123398999	None	0.6096657120978346	confidence_certainty, domain_disruption, domain_retention, recurrence	0.2927849588577119	1
NSD3	2	0.13333333333333333	0.01350896540516664	0.0226170123398999	None	0.6096657120978346	confidence_certainty, domain_disruption, domain_retention, recurrence	0.1280790765047707	2
FOXA1	1	0.06666666666666667	0.01350896540516664	0.0226170123398999	None	0.6096657120978346	confidence_certainty, domain_disruption, domain_retention, recurrence	0.10454966474006483	3
PAK1	1	0.06666666666666667	0.01350896540516664	0.0226170123398999	None	0.6096657120978346	confidence_certainty, domain_disruption, domain_retention, recurrence	0.10454966474006483	4
RAD51C	1	0.06666666666666667	0.01350896540516664	0.0226170123398999	None	0.6096657120978346	confidence_certainty, domain_disruption, domain_retention, recurrence	0.10454966474006483	5
VTCN1	1	0.06666666666666667	0.01350896540516664	0.0226170123398999	None	0.6096657120978346	confidence_certainty, domain_disruption, domain_retention, recurrence	0.10454966474006483	6

cutpoint_detection tables

cutpoint_scan

cutpoint	n_positive_at_or_below	n_positive_above	n_negative_at_or_below	n_negative_above	odds_ratio	p_value	neg_log10_p_value
1	2	8	2	0	0.0	0.09090909090909091	1.041392685158225

domain_disruption tables

frame_domain_contingency_table

5	5
1	1

frame_domain_contingency_tables_by_tumor_type

stratum	table
CUP	[1, 0], [1, 0]
IDC	[0, 2], [0, 0]
ILC	[0, 0], [0, 1]
LUAD	[4, 2], [0, 0]
PRAD	[0, 1], [0, 0]

domain_disruption_descriptives

Domain_key	Domain_name	Quantitative_call_count	Fully_retained_count	Truncated_count	Fully_lost_count	Mean_retained_fraction_among_non_retained_calls
auto_disruption_pf07679	Immunoglobulin I-set domain	12	2	0	10	0.0
auto_disruption_pf12661	Human growth factor-like EGF	0	0	0	0	None

domain_retention tables

frame_domain_contingency_table

6	4
0	2

frame_domain_contingency_tables_by_tumor_type

stratum	table
CUP	[2, 0], [0, 0]
IDC	[0, 0], [0, 2]
ILC	[0, 1], [0, 0]
LUAD	[4, 2], [0, 0]
PRAD	[0, 1], [0, 0]

domain_retention_descriptives

Domain_key	Domain_name	Quantitative_call_count	Fully_retained_count	Truncated_count	Fully_lost_count	Mean_retained_fraction_among_non_retained_calls
auto_key_domain	Neuregulin intracellular region	12	10	0	2	0.0

exon_retention tables

Exon_retention

Exon	Retained_event_count	Total_event_count	Retained_fraction
8	10	12	0.8333333333333334

frequency tables

Partner_gene_counts

Partner_gene	Event_count	Raw_event_count	Sample_count	Patient_count
CD74	9	9	9	7
FOXA1	1	1	1	1
NSD3	2	2	2	1
PAK1	1	1	1	1
RAD51C	1	1	1	1
VTCN1	1	1	1	1

genomic_position_recurrence tables

genomic_bin_recurrence

chromosome	bin_start	bin_end	n_events	event_ids
8	31994000	31995000	1	EVT-P-0030212-T01-IM6-15
8	32259000	32260000	1	EVT-P-0026615-T02-IM6-14
8	32309000	32310000	1	EVT-P-0008795-T01-IM5-5
8	32364000	32365000	1	EVT-P-0068238-T01-IM7-16
8	32505000	32506000	1	EVT-P-0028576-T01-IM6-12
8	32513000	32514000	1	EVT-P-0011180-T01-IM5-10
8	32546000	32547000	1	EVT-P-0010260-T01-IM5-9
8	32549000	32550000	1	EVT-P-0004752-T01-IM5-6
8	32565000	32566000	1	EVT-P-0058867-T01-IM7-2
8	32570000	32571000	1	EVT-P-0060522-T01-IM7-4
8	32574000	32575000	1	EVT-P-0058935-T02-IM7-3
8	32584000	32585000	1	EVT-P-0019346-T01-IM6-11

protein_position_genomic_spread

protein_position_aa	n_events	n_distinct_genomic_positions	genomic_span_bp	spans_multiple_chromosomes
168	8	8	78654	False
1	4	4	370123	False

junction_concentration tables

junction_counts

role	junction_exon	count	role_event_count	observed_share	random_breakage_share	excess_neg_log10_p
three_prime	6	8	8	1.0	0.5128331665290724	2.320191163159267

window_detection tables

window_scan

start_aa	end_aa	width_aa	n_events_inside	n_events_outside	n_positive_inside	n_positive_outside	odds_ratio	p_value	neg_log10_p_value	event_ids_inside
-199	1	200	4	8	2	8	0.0	0.09090909090909091	1.041392685158225	EVT-P-0008795-T01-IM5-5, EVT-P-0026615-T02-IM6-14, EVT-P-0030212-T01-IM6-15, EVT-P-0068238-T01-IM7-16
-99	1	100	4	8	2	8	0.0	0.09090909090909091	1.041392685158225	EVT-P-0008795-T01-IM5-5, EVT-P-0026615-T02-IM6-14, EVT-P-0030212-T01-IM6-15, EVT-P-0068238-T01-IM7-16
-49	1	50	4	8	2	8	0.0	0.09090909090909091	1.041392685158225	EVT-P-0008795-T01-IM5-5, EVT-P-0026615-T02-IM6-14, EVT-P-0030212-T01-IM6-15, EVT-P-0068238-T01-IM7-16
-24	1	25	4	8	2	8	0.0	0.09090909090909091	1.041392685158225	EVT-P-0008795-T01-IM5-5, EVT-P-0026615-T02-IM6-14, EVT-P-0030212-T01-IM6-15, EVT-P-0068238-T01-IM7-16
1	26	25	4	8	2	8	0.0	0.09090909090909091	1.041392685158225	EVT-P-0008795-T01-IM5-5, EVT-P-0026615-T02-IM6-14, EVT-P-0030212-T01-IM6-15, EVT-P-0068238-T01-IM7-16
1	51	50	4	8	2	8	0.0	0.09090909090909091	1.041392685158225	EVT-P-0008795-T01-IM5-5, EVT-P-0026615-T02-IM6-14, EVT-P-0030212-T01-IM6-15, EVT-P-0068238-T01-IM7-16
1	101	100	4	8	2	8	0.0	0.09090909090909091	1.041392685158225	EVT-P-0008795-T01-IM5-5, EVT-P-0026615-T02-IM6-14, EVT-P-0030212-T01-IM6-15, EVT-P-0068238-T01-IM7-16
68	168	100	8	4	8	2	None	0.09090909090909093	1.041392685158225	EVT-P-0004752-T01-IM5-6, EVT-P-0010260-T01-IM5-9, EVT-P-0011180-T01-IM5-10, EVT-P-0019346-T01-IM6-11, EVT-P-0028576-T01-IM6-12, ... (8 total)

start_aa	end_aa	width_aa	n_events_inside	n_events_outside	n_positive_inside	n_positive_outside	odds_ratio	p_value	neg_log10_p_value	event_ids_inside
118	168	50	8	4	8	2	None	0.09090909090909093	1.041392685158225	EVT-P-0004752-T01-IM5-6, EVT-P-0010260-T01-IM5-9, EVT-P-0011180-T01-IM5-10, EVT-P-0019346-T01-IM6-11, EVT-P-0028576-T01-IM6-12, ... (8 total)
143	168	25	8	4	8	2	None	0.09090909090909093	1.041392685158225	EVT-P-0004752-T01-IM5-6, EVT-P-0010260-T01-IM5-9, EVT-P-0011180-T01-IM5-10, EVT-P-0019346-T01-IM6-11, EVT-P-0028576-T01-IM6-12, ... (8 total)
168	193	25	8	4	8	2	None	0.09090909090909093	1.041392685158225	EVT-P-0004752-T01-IM5-6, EVT-P-0010260-T01-IM5-9, EVT-P-0011180-T01-IM5-10, EVT-P-0019346-T01-IM6-11, EVT-P-0028576-T01-IM6-12, ... (8 total)
168	218	50	8	4	8	2	None	0.09090909090909093	1.041392685158225	EVT-P-0004752-T01-IM5-6, EVT-P-0010260-T01-IM5-9, EVT-P-0011180-T01-IM5-10, EVT-P-0019346-T01-IM6-11, EVT-P-0028576-T01-IM6-12, ... (8 total)
168	268	100	8	4	8	2	None	0.09090909090909093	1.041392685158225	EVT-P-0004752-T01-IM5-6, EVT-P-0010260-T01-IM5-9, EVT-P-0011180-T01-IM5-10, EVT-P-0019346-T01-IM6-11, EVT-P-0028576-T01-IM6-12, ... (8 total)
168	368	200	8	4	8	2	None	0.09090909090909093	1.041392685158225	EVT-P-0004752-T01-IM5-6, EVT-P-0010260-T01-IM5-9, EVT-P-0011180-T01-IM5-10, EVT-P-0019346-T01-IM6-11, EVT-P-0028576-T01-IM6-12, ... (8 total)

top_windows

start_aa	end_aa	width_aa	n_events_inside	n_events_outside	n_positive_inside	n_positive_outside	odds_ratio	p_value	neg_log10_p_value	event_ids_inside
-24	1	25	4	8	2	8	0.0	0.09090909090909091	1.041392685158225	EVT-P-0008795-T01-IM5-5, EVT-P-0026615-T02-IM6-14, EVT-P-0030212-T01-IM6-15, EVT-P-0068238-T01-IM7-16
143	168	25	8	4	8	2	None	0.09090909090909093	1.041392685158225	EVT-P-0004752-T01-IM5-6, EVT-P-0010260-T01-IM5-9, EVT-P-0011180-T01-IM5-10, EVT-P-0019346-T01-IM6-11, EVT-P-0028576-T01-IM6-12, ... (8 total)

Per-event results (results.tsv)

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	domain_reconstitution	reconstitution_pfam_accession	reconstitution_five_prime_span	reconstitution_three_prime_span	reconstitution_junction_gap_fraction	reconstitution_junction_gap_residues	reconstitution_partner_breakpoint_protein_position	reconstitution_partner_domain_start_aa	reconstitution_partner_domain_end_aa	reconstitution_partner_breakpoint_mid_exon	reconstitution_detail	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-005886 7-T01-IM7-2	P-005886 7-T01-IM7	P-005886 7	Non-Small Cell Lung Cancer	LUAD	CD74-NRG1	CD74	in-frame	three_prime	32565439	6	168	True	168	True	exact match	retained	1.0	False													Neuregulin intracellular region	Immunoglobulin I-set domain			CD74 (NM_001025159) - NRG1 (NM_013956) rearrangement: t(5;8)(q32;p12)(chr5:g.149783529::chr8:g.32565439)| Protein Fusion: in frame {CD74:NRG1}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	domain_reconstitution	reconstitution_pfam_accession	reconstitution_five_prime_span	reconstitution_three_prime_span	reconstitution_junction_gap_fraction	reconstitution_junction_gap_residues	reconstitution_partner_breakpoint_protein_position	reconstitution_partner_domain_start_aa	reconstitution_partner_domain_end_aa	reconstitution_partner_breakpoint_mid_exon	reconstitution_detail	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-005893 5-T02-IM7-3	P-005893 5-T02-IM7	P-005893 5	Non-Small Cell Lung Cancer	LUAD	CD74-NRG1	CD74	in-frame	three_prime	32574604	6	168	True	168	True	exact match	retained	1.0	False													Neuregulin intracellular region	Immunoglobulin I-set domain			CD74 (NM_001025159) - NRG1 (NM_013956) fusion: t(5;8)(q32;p12)(chr5:g.149783327::chr8:g.32574604) | Protein Fusion: in frame {CD74:NRG1}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	domain_reconstitution	reconstitution_pfam_accession	reconstitution_five_prime_span	reconstitution_three_prime_span	reconstitution_junction_gap_fraction	reconstitution_junction_gap_residues	reconstitution_partner_breakpoint_protein_position	reconstitution_partner_domain_start_aa	reconstitution_partner_domain_end_aa	reconstitution_partner_breakpoint_mid_exon	reconstitution_detail	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-006052-T01-IM7-4	P-006052-T01-IM7	P-006052	Non-Small Cell Lung Cancer	LUAD	CD74-NRG1	CD74	in-frame	three_prime	32570123	6	168	True	168	True	exact match	retained	1.0	False													Neuregulin intracellular region	Immunoglobulin I-set domain		CD74 (NM_001025159) - NRG1 (NM_013956) rearrangement: t(5;8)(q32;p12)(chr5:g.149783541::chr8:g.32570123)| Protein Fusion: in frame {CD74:NRG1}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	domain_reconstitution	reconstitution_pfam_accession	reconstitution_five_prime_span	reconstitution_three_prime_span	reconstitution_junction_gap_fraction	reconstitution_junction_gap_residues	reconstitution_partner_breakpoint_protein_position	reconstitution_partner_domain_start_aa	reconstitution_partner_domain_end_aa	reconstitution_partner_breakpoint_mid_exon	reconstitution_detail	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-000879 5-T01-IM5-5	P-0008795-T01-IM5	P-0008795	Breast Cancer	IDC	FOXA1-NRG1	FOXA1	unknown	five_prime	32309025	1	1	True			breakpoint lies entirely upstream of the first coding exon (zero coding bases retained on this side); no residue to compare	lost	0.0	False														Neuregulin intracellular region; Immunoglobulin I-set domain		NRG1 (NM_013962) - FOXA1 (NM_004496) rearrangement : t(8;14)(p12;q21.1)(chr8:g.32309025::chr14:g.38061417) | Protein fusion: mid-exon (NRG1-FOXA1)	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	domain_reconstitution	reconstitution_pfam_accession	reconstitution_five_prime_span	reconstitution_three_prime_span	reconstitution_junction_gap_fraction	reconstitution_junction_gap_residues	reconstitution_partner_breakpoint_protein_position	reconstitution_partner_domain_start_aa	reconstitution_partner_domain_end_aa	reconstitution_partner_breakpoint_mid_exon	reconstitution_detail	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-00047522-T01-IM5-6	P-00047522-T01-IM5	P-00047522	Non-Small Cell Lung Cancer	LUAD	NRG1-CD74	CD74	in-frame	three_prime	32549502	6	168	True	168	True	exact match	retained	1.0	False													Neuregulin intracellular region	Immunoglobulin I-set domain			CD74 (NM_001025159) - NRG1 (NM_013956) rearrangement : t(5;8)(5q32;8p12)(chr8:g.32549502::chr5:g.149782966) | Protein fusion: in frame (CD74-NRG1)	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	domain_reconstitution	reconstitution_pfam_accession	reconstitution_five_prime_span	reconstitution_three_prime_span	reconstitution_junction_gap_fraction	reconstitution_junction_gap_residues	reconstitution_partner_breakpoint_protein_position	reconstitution_partner_domain_start_aa	reconstitution_partner_domain_end_aa	reconstitution_partner_breakpoint_mid_exon	reconstitution_detail	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-00047522-T02-IM5-7	P-00047522-T02-IM5	P-00047522	Non-Small Cell Lung Cancer	LUAD	NRG1-CD74	CD74	in-frame	three_prime	32549504	6	168	True	168	True	exact match	retained	1.0	False													Neuregulin intracellular region	Immunoglobulin I-set domain			CD74 (NM_001025159) - NRG1 (NM_013956) fusion (CD74 exons 1-6 fused with NRG1 exons 6-13) : t(5;8)(5q32;8p12)(chr5:g.149782964::chr8:g.32549504) | Protein Fusion: in frame {CD74:NRG1}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	domain_reconstitution	reconstitution_pfam_accession	reconstitution_five_prime_span	reconstitution_three_prime_span	reconstitution_junction_gap_fraction	reconstitution_junction_gap_residues	reconstitution_partner_breakpoint_protein_position	reconstitution_partner_domain_start_aa	reconstitution_partner_domain_end_aa	reconstitution_partner_breakpoint_mid_exon	reconstitution_detail	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-000475 2-T03-IM6-8	P-000475 2-T03-IM6	P-000475 2	Non-Small Cell Lung Cancer	LUAD	NRG1-CD74	CD74	in-frame	three_prime	32549504	6	168	True	168	True	exact match	retained	1.0	False													Neuregulin intracellular region	Immunoglobulin I-set domain		CD74 (NM_001025159) - NRG1 (NM_013956) fusion (CD74 exons 1-6 fused with NRG1 exons 6-13) : t(5;8)(5q32;8p12)(chr5:g.149782964::chr8:g.32549504) | Protein Fusion: in frame {CD74:NRG1}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	domain_reconstitution	reconstitution_pfam_accession	reconstitution_five_prime_span	reconstitution_three_prime_span	reconstitution_junction_gap_fraction	reconstitution_junction_gap_residues	reconstitution_partner_breakpoint_protein_position	reconstitution_partner_domain_start_aa	reconstitution_partner_domain_end_aa	reconstitution_partner_breakpoint_mid_exon	reconstitution_detail	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-001026 0-T01-IM5-9	P-001026 0-T01-IM5	P-001026 0	Cancer of Unknown Primary	CUP	NRG1-CD74	CD74	in-frame	three_prime	32546211	6	168	True	168	True	exact match	retained	1.0	False													Neuregulin intracellular region	Immunoglobulin I-set domain		CD74 (NM_001025159) - NRG1 (NM_013956) fusion (CD74 exons 1-6 fused in frame with NRG1 exons 3-13): t(5;8)(q32;p12)(chr5:g.149783390::chr8:g.32546211)| Protein fusion: in frame (CD74-NRG1)	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	domain_reconstitution	reconstitution_pfam_accession	reconstitution_five_prime_span	reconstitution_three_prime_span	reconstitution_junction_gap_fraction	reconstitution_junction_gap_residues	reconstitution_partner_breakpoint_protein_position	reconstitution_partner_domain_start_aa	reconstitution_partner_domain_end_aa	reconstitution_partner_breakpoint_mid_exon	reconstitution_detail	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-001118 0-T01-IM5-10	P-001118 0-T01-IM5	P-001118 0	Non-Small Cell Lung Cancer	LUAD	NRG1-CD74	CD74	unknown	three_prime	32513214	6	168	True	168	True	exact match	retained	1.0	False													Neuregulin intracellular region	Immunoglobulin I-set domain			CD74 (NM_001025159) - NRG1 (NM_013956) rearrangement: t(5;8)(q32;p12)(chr5:g.149782758::chr8:g.32513214)| Protein fusion: mid-exon (CD74-NRG1)	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	domain_reconstitution	reconstitution_pfam_accession	reconstitution_five_prime_span	reconstitution_three_prime_span	reconstitution_junction_gap_fraction	reconstitution_junction_gap_residues	reconstitution_partner_breakpoint_protein_position	reconstitution_partner_domain_start_aa	reconstitution_partner_domain_end_aa	reconstitution_partner_breakpoint_mid_exon	reconstitution_detail	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-001934 6-T01-IM6-11	P-001934 6-T01-IM6	P-001934 6	Non-Small Cell Lung Cancer	LUAD	NRG1-CD74	CD74	unknown	three_prime	32584466	6	168	True	168	True	exact match	retained	1.0	False													Neuregulin intracellular region	Immunoglobulin I-set domain		CD74 (NM_001025159) - NRG1 (NM_013956) fusion (CD74 exons 1-7 fused with NRG1 exons 6-13) : t(5;8)(q32;p12) (chr5:g.149782781::chr8:g.32584466) | Protein Fusion : mid-exon {CD74:NRG1}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	domain_reconstitution	reconstitution_pfam_accession	reconstitution_five_prime_span	reconstitution_three_prime_span	reconstitution_junction_gap_fraction	reconstitution_junction_gap_residues	reconstitution_partner_breakpoint_protein_position	reconstitution_partner_domain_start_aa	reconstitution_partner_domain_end_aa	reconstitution_partner_breakpoint_mid_exon	reconstitution_detail	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-002857 6-T01-IM6-12	P-002857 6-T01-IM6	P-002857 6	Prostate Cancer	PRAD	NRG1-NSD3	NSD3	unknown	three_prime	32505812	6	168	True	168	True	exact match	retained	1.0	False													Neuregulin intracellular region	Immunoglobulin I-set domain		WHSC1L1 (NM_023034) rearrangement: c.2758+79: WHSC1L1_chr8:g.32505812del | Antisense Fusion	NA
EVT-P-002857 6-T02-IM6-13	P-002857 6-T02-IM6	P-002857 6	Prostate Cancer	PRAD	NRG1-NSD3	NSD3	unknown	three_prime	32505812	6	168	True	168	True	exact match	retained	1.0	False													Neuregulin intracellular region	Immunoglobulin I-set domain		WHSC1L1 (NM_023034) rearrangement: c.2758+79: WHSC1L1_chr8:g.32505812del | Antisense Fusion	NA

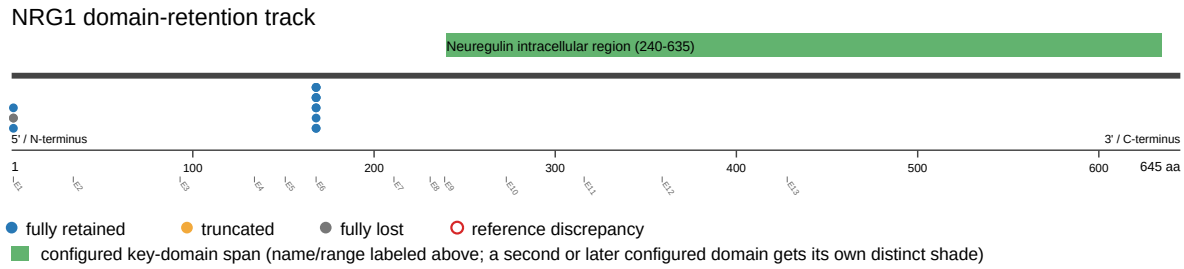
event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	domain_reconstitution	reconstitution_pfam_accession	reconstitution_five_prime_span	reconstitution_three_prime_span	reconstitution_junction_gap_fraction	reconstitution_junction_gap_residues	reconstitution_partner_breakpoint_protein_position	reconstitution_partner_domain_start_aa	reconstitution_partner_domain_end_aa	reconstitution_partner_breakpoint_mid_exon	reconstitution_detail	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-002661 5-T02-IM6-14	P-002661 5-T02-IM6	P-002661 5	Breast Cancer	ILC	PAK1-NRG1	PAK1	unknown	three_prime	32259693	1	1	True	1	True	exact match	retained	1.0	False												Neuregulin intracellular region; Immunoglobulin I-set domain			PAK1 (NM_002576) Rearrangement: t(8;11)(p12;q13.5)(chr8:g.32259693::chr11:g.77036046) | Antisense Fusion	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	domain_reconstitution	reconstitution_pfam_accession	reconstitution_five_prime_span	reconstitution_three_prime_span	reconstitution_junction_gap_fraction	reconstitution_junction_gap_residues	reconstitution_partner_breakpoint_protein_position	reconstitution_partner_domain_start_aa	reconstitution_partner_domain_end_aa	reconstitution_partner_breakpoint_mid_exon	reconstitution_detail	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-003012-T01-IM6-15	P-003012-T01-IM6	P-003012	Breast Cancer	IDC	RAD51C-NRG1	RAD51C	unknown	five_prime	32994096	1	1	True			breakpoint lies entirely upstream of the first coding exon (zero coding bases retained on this side); no residue to compare	lost	0.0	False														Neuregulin intracellular region; Immunoglobulin I-set domain		NRG1 (NM_013956) - RAD51C (NM_058216) rearrangement: t(8;17)(p12;q22)(chr8:g.32994096::chr17:g.56787249) | Protein Fusion: mid-exon {NRG1:RAD51C}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	domain_reconstitution	reconstitution_pfam_accession	reconstitution_five_prime_span	reconstitution_three_prime_span	reconstitution_junction_gap_fraction	reconstitution_junction_gap_residues	reconstitution_partner_breakpoint_protein_position	reconstitution_partner_domain_start_aa	reconstitution_partner_domain_end_aa	reconstitution_partner_breakpoint_mid_exon	reconstitution_detail	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-006823 8-T01-IM7-16	P-006823 8-T01-IM7	P-006823 8	Cancer of Unknown Primary	CUP	VTCN1-NRG1	VTCN1	in-frame	three_prime	32364219	1	1	True	1	True	exact match	retained	1.0	False												Neuregulin intracellular region; Immunoglobulin I-set domain			VTCN1 (NM_024626) - NRG1 (NM_013956) rearrangement: t(1;8)(p13.1;p12)(chr1:g.117699192::chr8:g.32364219) | Protein Fusion: in frame {VTCN1:NRG1}	NA

Figures

domain retention outliers



fusion schematic

